R823-HG-T006

Del classe al modul: §4, theorem 2

Status: provisional construction test. This is a source-keyed translation experiment, not a native-reviewed standard, not an intelligibility result, and not a completed translation of the R823 volume.

Source boundary. German cumulative authority, lines 21209–21254. Authority SHA-256 begins `EE8955BCF7A2639`; exact slice SHA-256 begins `2F90BDA5829FDABF`. Every translated source segment is aligned in the accompanying clause map. Line 21255 is blank; the next semantic source unit begins at line 21256 with Chapter II.

§ 4 (continuation): Le construction converse

2. *Cata classe de representation es generate, in le maniera indicate sub 1, per un modul de representation. (Le scope es de novo limitate al representationes direct.)*

Demostracio. Nos prende un representation determinate del classe, date per le association homomorf $c \mapsto C$. Si n es le grado del representation, nos forma con n indeterminates $x_1, \dots, x_n$ un modul de T con scalars al dextra:

$$\mathfrak{M} = x_1\mathsf{T} + \cdots + x_n\mathsf{T}.$$

Su reglas de calculation es

$$\begin{gathered} \sum x_i\tau_i + \sum x_i\overline{\tau}_i = \sum x_i(\tau_i + \overline{\tau}_i), \\ \sum x_i\tau_i = \sum x_i\overline{\tau}_i \quad \iff \quad \tau_i = \overline{\tau}_i \text{ pro cata } i, \\ \left(\sum x_i\tau_i\right)\varrho = \sum x_i\tau_i\varrho. \end{gathered}$$

Le secunde equalitate vale *si e solmente si* le coefficientes correspondente es equal pro cata i. Nos adde a illo un action de $\mathfrak{o}$ scripte al sinistra per le definition

$$c\sum x_i\tau_i = \sum x_i'\tau_i, \qquad (x_1', \dots, x_n') = (x_1, \dots, x_n)C,$$

ubi C significa le matrice del representation associate al elemento c de $\mathfrak{o}$.

Iste definition rende $\mathfrak{M}$ un $\mathfrak{o}, \mathsf{T}$-bimodul. In prime loco, del relationes homomorf

$$c + d \mapsto C + D \quad \text{e} \quad cd \mapsto CD$$

se deriva

$$(c + d)x_i = cx_i + dx_i$$

e

$$cd \cdot x_i = c \cdot dx_i,$$

$$c\left(\sum x_i\tau_i + \sum x_i\overline{\tau}_i\right) = c\sum x_i\tau_i + c\sum x_i\overline{\tau}_i.$$

Isto es le proprietates del modul con action scripte al sinistra.

In secunde loco, se ha evidentemente

$$c\left(\sum x_i\tau_i\varrho\right) = \left(c\sum x_i\tau_i\right) \cdot \varrho,$$

que es le lege associativ mixed.

Assi $\mathfrak{M}$ es un modul de representation, e le representation usate es generate per illo in le maniera indicate sub 1, quando le base $x_1, \ldots, x_n$ es usate pro realisar su transformationes linear. Le altere bases dona le restante representationes del classe.

Pro le representationes reciproc e le moduls reciproc de representation, toto procede in le mesme maniera. Resulta que le modul

$$\mathfrak{M}^* = \mathsf{T}^* x_1^* + \cdots + \mathsf{T}^* x_n^*$$

genera un representation reciproc de $\mathfrak{o}$ in T^* per associar al elemento c de $\mathfrak{o}$ le matrice C in le equation

$$c\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} cx_1 \\ \vdots \\ cx_n \end{pmatrix} = C\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Le bases differente dona de novo le representationes differente de un classe.

Nota editorial de senso

Le parola candidate *demostracio* nomina un prova mathematic formal; illo non nomina un test o experimento. Le theorem es constructiv: illo parte de un representation fixe e construe un modul que la genera. Illo non affirma que le modul es unic.

Le latere del duo actiones es declarate per le ordine del formulas: $x_i\tau_i$ pone T al dextra, durante que $c\sum x_i\tau_i$ pone $\mathfrak{o}$ al sinistra. Le texto non depende de un parola isolate pro “left” o “right”. Le equivalence del summas afirma unicitate del coefficientes in le base formal; illo non affirma que T es commutativ.

Le relationes $c + d \mapsto C + D$ e $cd \mapsto CD$ es retenite in lor ordine. Le *lege associativ mixed* es exactemente le compatibilitate

$$c(m\varrho) = (cm)\varrho;$$

illo non es un affirmation que $\mathfrak{o}$ e T commuta como anels.

In le caso direct, le base es un tupla-riga e C acta al dextra. In le caso reciproc, le fonte usa un columna e C acta al sinistra. Iste cambio de orientation es conservate; le asteriscos de $\mathfrak{M}^*$, T^* e x_i^* non es eliminate ni transferite silentiose.

Limites del tranche

Le grammatica e le vocabulario es candidatos de construction. Nulle affirmation de comprehension human, de validation native, o de intelligibilitate marginal es facite. Il ha zero observationes human e nulle resultato pilot. Le proxime unitate semantic comencia al linea 21256 con le titulo de Capitulo II; §5 comencia al linea 21260.